Ch1

Matrix Operations & Calculations

For a 2×2 matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$:

• Determinant (2×2): $|A| = a_{11}a_{22} - a_{21}a_{12}$. • Inverse ($2 \times 2$): $A^{-1} = \frac{1}{|A|}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$. • **Note**: A^{-1} exists only if $|A| \neq 0$

• If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ $\quad A^{-1} == \frac{1}{|A|}\begin{bmatrix} a_{22}a_{33} - a_{32}a_{23} & a_{32}a_{13} - a_{12}a_{33} & a_{12}a_{23} - a_{22}a_{13} \\ a_{31}a_{23} - a_{21}a_{33} & a_{11}a_{33} - a_{31}a_{13} & a_{21}a_{13} - a_{11}a_{23} \\ a_{21}a_{32} - a_{31}a_{22} & a_{31}a_{12} - a_{11}a_{32} & a_{11}a_{22} - a_{21}a_{12} \end{bmatrix}$

• Trace: $tr(A) = \sum_{i=1}^{p} a_{ii}$. • **Note**: Cyclic property holds, meaning $tr(AB) = tr(BA)$ and $tr(ABC) = tr(CAB) = tr(BCA)$, provided the dimensions allow for computation. $tr(A + B) = tr(A) + tr(B)$

• Positive Definite & Semidefinite:

  • A symmetric matrix A is positive semidefinite ($A \geq 0$) if $x'Ax \geq 0$ for any nonzero vector x.

  • A symmetric matrix A is positive definite ($A > 0$) if $x'Ax > 0$ for any nonzero vector x.

• Eigenvalues (λ_i): Calculated by solving the characteristic equation $|A - \lambda I_p| = 0$.

• Eigenvectors (h_i): Solved via $Ah_i = \lambda_i h_i$. A normalized eigenvector requires $h_i'h_i = 1$.

  • **Note**: For a real symmetric matrix, eigenvalues are real, and eigenvectors corresponding to distinct eigenvalues are strictly orthogonal ($h_i'h_j = 0$).

• Spectral Decomposition: $A = HDH'$. $A^{1/2} = HD^{1/2}H'$

  • **Note**: $D = diag(\lambda_1, \ldots, \lambda_p)$ and $H = (h_1 \ldots h_p)$ consists of normalized eigenvectors. Because H is orthogonal, $H'H = I_p$.

• Symmetric Square Root: $A^{1/2} = HD^{1/2}H'$, where $D^{1/2} = diag(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_p})$. D是只保留Σ主对角线方差的对角阵，作用是将协方差标准化为[-1,1]之间的相关系数

  • **Note**: This is only computable if A is positive semidefinite ($A \geq 0$), guaranteeing all $\lambda_i \geq 0$.

Population Random Vectors

• Population Covariance Matrix: $\Sigma = Var(x) = E((x - E(x))(x - E(x))') = E(xx') - E(x)E(x)'$.

  • **Note**: Σ is symmetric and positive semidefinite ($\Sigma \geq 0$). Diagonal element σ_{ii} is the variance of X_i, off-diagonal σ_{ij} is the covariance between X_i and X_j.

• Covariance Between Two Vectors (x and y): $Cov(x, y) = E((x - E(x))(y - E(y))') = E(xy') - E(x)E(y)'$.

• Population Correlation Matrix: $R = D^{-1/2}\Sigma D^{-1/2}$.

  • **Note**: D contains the variances, so $D^{-1/2} = diag(\frac{1}{\sqrt{\sigma_{11}}}, \ldots, \frac{1}{\sqrt{\sigma_{pp}}})$.

• Linear Combinations (Vectors): $E(a'x) = a'E(x)$. $Var(a'x) = a'\Sigma a$. $Cov(a'x, b'x) = a'\Sigma b$.

• Linear Combinations (Matrices): $Var(Ax) = A\Sigma A'$. $Cov(Ax, Bx) = A\Sigma B'$.

• **Note**: $a'x$ outputs a scalar variance, while Ax outputs a covariance matrix.

Sample Data Matrix & Estimators

• Sample Mean Vector: $\overline{x} = \frac{1}{n}X'1_n$. • **Note**: It is an unbiased estimator, $E(\overline{x}) = \mu$.

Its theoretical variance is $Var(\overline{x}) = \frac{1}{n}\Sigma$. (Proof: $Var(\frac{1}{n}\sum x_i) = \frac{1}{n^2}\sum Var(x_i) = \frac{1}{n}\Sigma$).

• SSCP Matrix: $A = X'X - n\overline{x}\overline{x}' = (X - 1_n\overline{x}')'(X - 1_n\overline{x}')$.

  • Element Calculation: $a_{ij} = \sum_{k=1}^{n}(x_{ki} - \overline{x}_i)(x_{kj} - \overline{x}_j)$.

  • Positive Semidefinite: $A \geq 0$. (Proof: For any $y = Bx$, $x'B'Bx = y'y \geq 0$, and A is in the form of $B'B$).

• Sample Covariance Matrix: $S = \frac{1}{n-1}A = \frac{1}{n-1}\sum_{k=1}^{n}(x_k - \overline{x})(x_k - \overline{x})'$.

  • **Note**: The denominator must strictly be $n - 1$ for S to be an unbiased estimator. $E(S) = \Sigma$. (Proof: $E(A) = (n - 1)\Sigma$, hence $E(S) = \Sigma$). The matrix S is always symmetric positive semidefinite.

• Sample Correlation Matrix: $R = D^{-1/2}SD^{-1/2}$.

  • **Note**: Here, $D^{-1/2} = diag(\frac{1}{\sqrt{s_{11}}}, \ldots, \frac{1}{\sqrt{s_{pp}}})$ relies on the sample variances s_{ii} found on the diagonal of S.

Key Univariate Distributions (Building Blocks)

1. Chi-square: If $Z_1, \ldots, Z_\nu$ are iid $N(0, 1)$, then $X = Z_1^2 + \ldots + Z_\nu^2 \sim \chi_\nu^2$.

  • Sample Variance Relation: $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

2. Student's t: $T = \frac{Z}{\sqrt{X/\nu}} \sim t_\nu$, provided $Z \sim N(0,1)$ and $X \sim \chi^2_\nu$ are independent.

- **Note**: For sample testing, if $Y_1, \ldots, Y_n \sim iid\ N(\mu, \sigma^2)$, then the sample mean is distributed as $\overline{Y} \sim N(\mu, \frac{\sigma^2}{n})$. This leads to the test statistic $T = \frac{\sqrt{n}(\overline{Y}-\mu)}{S} \sim t_{n-1}$.

3. Fisher's F: $F = \frac{X_1/u}{X_2/v} \sim F_{u,v}$, provided $X_1 \sim \chi^2_u$ and $X_2 \sim \chi^2_v$ are independent.

- Variance Ratio: If $X_1, \ldots, X_m \sim iid\ N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_n \sim iid\ N(\mu_2, \sigma_2^2)$ are two independent samples , then $F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F_{m-1,n-1}$.

- **Note**: If a variable $T \sim t_v$, then its square $T^2 \sim F_{1,v}$.

Ch2

Notation (Multivariate Normal)

- $N_p(\mu, \Sigma)$: p-variate normal distribution with mean vector μ and covariance matrix Σ.
- d_i^2: Squared generalized distance (Mahalanobis distance) for the i-th observation, defined as $d_i^2 = (x_i - \overline{x})' S^{-1} (x_i - \overline{x})$.
- A: Sum of Squares and Cross product (SSCP) matrix, defined as $\sum_{i=1}^n (x_i - \overline{x})(x_i - \overline{x})'$.
- $W_p(m, \Sigma)$: Wishart distribution with m degrees of freedom and scale matrix Σ.
- $\Sigma_{11\bullet2}$: Conditional covariance matrix of x_1 given x_2, defined as $\Sigma_{11\bullet2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

The Multivariate Normal Distribution

- Density Function: $f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\}$.
- Bivariate Normal ($p=2$): Covariance matrix is $\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$.
- Inverse of Covariance: $\Sigma^{-1} = \dfrac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$.
- **Note**: The density $f(x_1, x_2)$ relies heavily on the correlation coefficient ρ between X_1 and X_2.

Properties & Linear Transformations

- Linear Combination: If $x \sim N_p(\mu, \Sigma)$ and $a \neq 0$, then $a'x \sim N(a'\mu, a'\Sigma a)$.
- Addition: If $x \sim N_p(\mu_1, \Sigma_1)$ and $y \sim N_p(\mu_2, \Sigma_2)$ are independent, then $x + y \sim N_p(\mu_1 + \mu_2, \Sigma_1 + \Sigma_2)$.
- Matrix Transformation: For a $q \times p$ matrix B and $q \times 1$ vector b, $y = Bx + b \sim N_q(B\mu + b, B\Sigma B')$.

Random Samples & MLE

Let $x_1, \ldots, x_n$ be iid $N_p(\mu, \Sigma)$: Joint Likelihood Function:

$L(\mu, \Sigma | x_1, \ldots, x_n) = (2\pi)^{-np/2}|\Sigma|^{-n/2} \exp\{-\frac{1}{2}\sum_{i=1}^n (x_i-\mu)'\Sigma^{-1}(x_i-\mu)\} = (2\pi)^{-np/2}|\Sigma|^{-n/2} \exp\{-\frac{1}{2}tr(\Sigma^{-1}A)\} \exp\{-\frac{n}{2}(\overline{x}-\mu)'\Sigma^{-1}(\overline{x}-\mu)\}$

- Sample Mean Distribution: $\overline{x} \sim N_p(\mu, \frac{1}{n}\Sigma)$
- Quadratic Form of Mean: $n(\overline{x}-\mu)'\Sigma^{-1}(\overline{x}-\mu) \sim \chi^2_p$.
- Mahalanobis Distance Distribution: $d_i^2 = (x_i - \overline{x})' S^{-1}(x_i - \overline{x}) \sim \chi^2_p$ asymptotically for large n.
- MLE of μ: $\hat{\mu} = \overline{x}$. • MLE of Σ: $\hat{\Sigma} = \frac{1}{n}A = \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})(x_i - \overline{x})'$.
- Unbiased Estimator of Σ: $S = \frac{1}{n-1}A$. **Note**: $\overline{x}$ and S are independent.

Marginal and Conditional Distributions

Partition $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$, $\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$:

1. Marginal Distribution: $x_1 \sim N_q(\mu_1, \Sigma_{11})$. Any subset of x is normally distributed.从x中任意取出q个分量组成的子集，这个子集的边缘分布也必然服从q元正态分布

2. Independence: x_1 and x_2 are independent if and only if $\Sigma_{12} = 0$.

3. (Schur Complement) Let $\Sigma_{11\bullet2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$, we have: $x_1 - \Sigma_{12}\Sigma_{22}^{-1}x_2 \sim N_q(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11\bullet2})$.

Note: This residual vector is strictly independent of x_2. 目的是构造一个与x2完全独立的新rm

4. Conditional Distribution: The distribution of x_1 given $x_2 = x_2^0$ is $N_q(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2^0 - \mu_2), \Sigma_{11\bullet2})$.

Wishart Distribution & Quadratic Forms

- Wishart Definition: If $z_1, \ldots, z_m \sim iid\ N_p(0, \Sigma)$, then $B = \sum_{i=1}^m z_i z_i' \sim W_p(m, \Sigma)$.
  - Density Function (B): $f_m(B|\Sigma) = \frac{|B|^{(m-p-1)/2}|\Sigma|^{-m/2}e^{-tr(\Sigma^{-1}B)/2}}{K}$, where $K = 2^{mp/2}\pi^{p(p-1)/4}\prod_{i=1}^p \Gamma(\frac{m+1-i}{2})$.

• SSCP Distribution: The matrix $A = (n-1)S \sim W_p(n-1, \Sigma)$.

• Density Function (A): $f_{n-1}(A|\Sigma) = \frac{|A|^{(n-p-2)/2}|\Sigma|^{-(n-1)/2}e^{-tr(\Sigma^{-1}A)/2}}{K}$, where $K = 2^{(n-1)p/2}\pi^{p(p-1)/4}\prod_{i=1}^{p}\Gamma(\frac{n-i}{2})$.

• Connection between Wishart and χ^2: In the univariate case ($p = 1$), $B = Z_1^2 + \ldots + Z_m^2$ where $Z_1, \ldots, Z_m \sim iid\ N(0, \sigma^2)$.

• The density of B is $f_m(B|\sigma^2) = \frac{B^{m/2-1}(\sigma^2)^{-m/2}e^{-B/(2\sigma^2)}}{K}$, where $K = 2^{m/2}\Gamma(m/2)$.

• We may write $B \sim W_1(m, \sigma^2)$ or $B/\sigma^2 \sim \chi^2_m$. • If we let $V = B/\sigma^2$, the density function of V is $f_V(v) = \frac{v^{m/2-1}e^{-v/2}}{2^{m/2}\Gamma(m/2)}$.

• Cochran's Theorem: Let $Y_1, \ldots, Y_n \sim iid\ N(0, \sigma^2)$. If $\sum_{i=1}^{n} Y_i^2 = Q_1 + \ldots + Q_k$ where $Q_i = y'A_iy$ and $\sum rank(A_i) = n$, then Q_i are independent and $Q_i/\sigma^2 \sim \chi^2_{rank(A_i)}$.

• Theorem 3 (Special Case): If $Y_1, \ldots, Y_n \sim iid\ N(\mu, \sigma^2)$, then $\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}\frac{(Y_i - \overline{Y})^2}{\sigma^2} \sim \chi^2_{n-1}$.

• Lemma 4: Let G be a $p \times p$ matrix such that $G^2 = G$ and $tr(G) = k$. Then there exists a $k \times p$ matrix E such that $EE' = I_k$ and $E'E = G$.

• General Quadratic Form Application: If $x \sim N_p(\mu, \Sigma)$, then $y = \Sigma^{-1/2}(x - \mu) \sim N_p(0, I_p)$, which implies $y'y = (x-\mu)'\Sigma^{-1}(x-\mu) \sim \chi^2_p$.

Practical Procedures: Outliers, Generation & Testing

• Generating Normal Vectors: Generate independent $z_i \sim N_p(0, I_p)$. Use spectral decomposition $\Sigma = HDH'$, then compute $x_i = \sigma z_i + \mu = HD^{1/2}z_i + \mu$ to get $x_i \sim N_p(\mu, \Sigma)$. rescaling the original spherical shape of zi to an elliptical shape and then rotating the axis along the direction of the eigenvectors.

• Outlier Detection:

1. Compute Mahalanobis distance $d_i^2 = (x_i - \overline{x})'S^{-1}(x_i - \overline{x})$ for each observation.
2. An observation is flagged as a potential outlier if $d_i^2 > q_p(0.99)$, where $q_p(0.99)$ is the 99th percentile of the χ^2_p distribution.

• Checking Multivariate Normality (QQ-Plot):

1. Compute d_i^2 for all observations.
2. Order them ascendingly as $d^2_{(i)}$.
3. Compute theoretical quantiles $q_p(i)$ from χ^2_p using probability $(i - 0.5)/n$.
4. Plot $d^2_{(i)}$ against $q_p(i)$. A straight line suggests multivariate normality.
5. Kolmogorov-Smirnov (KS) Test: Use test statistic $D_n = \sup_x |F_n(x) - F(x)|$ to formally test if d_i^2 follows χ^2_p.

Ch3

Notation (Hypothesis Testing)

• d_j: The difference vector for the j-th pair in a paired comparison, defined as $x_{1j} - x_{2j}$.

• $\overline{d}$: Sample mean vector of the differences.

• S_d: Sample covariance matrix of the differences.

• S_p: Pooled sample covariance matrix for two independent samples.

• C: A $q \times p$ contrast matrix of constants used for linear combinations.

Hotelling's T^2 Definition & Foundation

• Definition: If d and M are independently distributed as $d \sim N_p(0, I)$ and $M \sim W_p(m, I)$, then $md'M^{-1}d \sim T^2(p, m)$.

• Key Results:

1. If $d \sim N_p(\mu, \Sigma)$ and $M \sim W_p(m, \Sigma)$ are independent, then $m(d-\mu)'M^{-1}(d-\mu) \sim T^2(p, m)$.
2. Relation to F-distribution: $\frac{m-p+1}{p}\frac{T^2(p,m)}{m} = F_{p,m-p+1}$.

1. One-Sample Test for Mean Vector ($\mu = \mu_0$)

Case A: Population Covariance Σ is KNOWN

• Test Statistic: $n(\overline{x} - \mu_0)'\Sigma^{-1}(\overline{x} - \mu_0) \sim \chi^2_p$.

• Decision Rule: Reject H_0 if the statistic $> \chi^2_p(\alpha)$.

• **Note**: This is the exact multivariate analog of the univariate Z-test. Since Σ is known, we use the Chi-square distribution directly.

Case B: Population Covariance Σ is UNKNOWN (Hotelling's T^2)

• Test Statistic Definition: $T_0^2 = n(\overline{x} - \mu_0)'S^{-1}(\overline{x} - \mu_0)$.

• Distribution: $T_0^2 \sim T^2(p, n-1)$.

• **Note**: Here p refers to the dimension (number of variables) of the random vector x.

- F-Transformation: $\frac{n-p}{p}\frac{T_0^2}{n-1} \sim F_{p,n-p}$.
- Decision Rule: Reject H_0 if $\frac{n-p}{p}\frac{T_0^2}{n-1} > F_{p,n-p}(\alpha)$.
  - **Note**: This is the multivariate analog of the one-sample t-test. S must be the unbiased estimator with denominator $n-1$. Pay close attention to the degrees of freedom for the F-distribution: numerator is p, denominator is $n-p$.

2. Paired Comparison Test

Used when experimental units are paired or measurements are taken twice on the same unit (dependent samples).

- Assumption: $d_1, \ldots, d_n \sim iid\ N_p(\delta, \Sigma)$. difference e.g. $d_1 = x_{11} - x_{21}$
- Hypothesis: $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$, where δ is the population mean difference.
- Difference Vector: $d_j = x_{1j} - x_{2j}$ for $j = 1, \ldots, n$.
- Sample Difference Mean: $\overline{d} = \frac{1}{n}\sum_{j=1}^{n} d_j$.
- Sample Difference Covariance: $S_d = \frac{1}{n-1}\sum_{j=1}^{n}(d_j - \overline{d})(d_j - \overline{d})'$.
- Test Statistic: $T^2 = n\overline{d}' S_d^{-1}\overline{d}$.
- Transformation & Rule: Reject H_0 if $\frac{n-p}{p}\frac{T^2}{n-1} = \frac{n-p}{p}\frac{n}{n-1}(\overline{d} - \delta)' S_d^{-1}(\overline{d} - \delta) > F_{p,n-p}(\alpha)$.
  - **Note**: Paired comparison reduces a two-sample problem into a one-sample T^2 test on the differences d_j. The dimensions p and sample size n refer to the difference vectors, not the original separate measurements.

3. Two-Sample T^2 Test (Independent Samples)

Used to test $H_0 : \mu_1 = \mu_2$ for two independent groups. 前提假设：两个总体的协方差矩阵必须相等

- Assumption: Both populations must share the same covariance matrix Σ.
- Pooled Covariance: $S_p = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$.
  - **Note**: This explicitly uses the denominator $(n_1 + n_2 - 2)$ to ensure S_p is an unbiased estimator of the common Σ.
- Test Statistic: $T^2 = \frac{n_1 n_2}{n_1+n_2}(\overline{x}_1 - \overline{x}_2)' S_p^{-1}(\overline{x}_1 - \overline{x}_2)$.
- Transformation & Rule: Reject H_0 if $\frac{n_1+n_2-p-1}{p}\frac{T^2}{n_1+n_2-2} > F_{p,n_1+n_2-p-1}(\alpha)$.
  - **Note**: Notice the scaling factor $\frac{n_1 n_2}{n_1+n_2}$, which comes from the variance of $(\overline{x}_1 - \overline{x}_2)$. The F-distribution denominator degrees of freedom change to $n_1 + n_2 - p - 1$.

4. Testing Linear Combinations & Repeated Measurements

Used to test relationships between variables, such as "is the mean of variable 1 equal to variable 2?".

- General Hypothesis: $H_0 : C\mu = 0$ where C is a $q \times p$ constant matrix.
- Under H_0: $C\overline{x} \sim N_q(0, \frac{1}{n}C\Sigma C')$.
- Test Statistic: $T^2 = n\overline{x}' C'(CSC')^{-1}C\overline{x}$.
- Transformation & Rule: Reject H_0 if $\frac{n-q}{q}\frac{T^2}{n-1} > F_{q,n-q}(\alpha)$.
  - **Note**: We replace p with q (the number of rows in C) in the F-transformation formula because applying C reduces the dimensionality from p down to q.

Application: Repeated Measurements

- Goal: Test if means over p time periods are identical ($H_0 : \mu_1 = \mu_2 = \ldots = \mu_p$).
- Contrast Matrix C: A $(p-1) \times p$ matrix constructed as adjacent differences (e.g., row 1 is $1, -1, 0..$; row 2 is $0, 1, -1..$).
- Rule: Since $q = p - 1$, the test rejects if $\frac{n-p+1}{p-1}\frac{T^2}{n-1} > F_{p-1,n-p+1}(\alpha)$.

Important Proofs & Derivations

Unbiasedness of Sample Covariance ($E(S) = \Sigma$)

- First, expand the SSCP matrix $A = \sum_{i=1}^{n} x_i x_i' - n\overline{x}\overline{x}'$.
- Take the expectation: $E(A) = \sum_{i=1}^{n} E(x_i x_i') - nE(\overline{x}\overline{x}')$.
- Using the property $E(yy') = Var(y) + E(y)E(y)'$, we get $E(x_i x_i') = \Sigma + \mu\mu'$.
- Similarly, $E(\overline{x}\overline{x}') = Var(\overline{x}) + E(\overline{x})E(\overline{x})' = \frac{1}{n}\Sigma + \mu\mu'$.
- Substitute these back: $E(A) = n(\Sigma + \mu\mu') - n(\frac{1}{n}\Sigma + \mu\mu') = n\Sigma + n\mu\mu' - \Sigma - n\mu\mu' = (n-1)\Sigma$.
- Since $S = \frac{1}{n-1}A$, it follows that $E(S) = \frac{1}{n-1}(n-1)\Sigma = \Sigma$.

Maximum Likelihood Estimator (MLE) of Σ using the Trace Trick

- The log-likelihood function involves the sum: $\sum_{i=1}^{n}(x_i - \mu)'\Sigma^{-1}(x_i - \mu)$.
- Since a scalar equals its trace, apply the cyclic property $tr(AB) = tr(BA)$: $\sum_{i=1}^{n} tr((x_i - \mu)'\Sigma^{-1}(x_i - \mu)) = \sum_{i=1}^{n} tr(\Sigma^{-1}(x_i - \mu)(x_i - \mu)')$.
- Move the summation inside the trace: $tr(\Sigma^{-1}\sum_{i=1}^{n}(x_i - \mu)(x_i - \mu)')$.
- By expanding $(x_i - \mu) = (x_i - \overline{x}) + (\overline{x} - \mu)$ and cross-multiplying, the cross terms sum to zero. The term becomes $tr(\Sigma^{-1}A) + n(\overline{x} - \mu)'\Sigma^{-1}(\overline{x} - \mu)$.
- Maximizing the likelihood with respect to μ forces the second term to 0 (hence $\hat{\mu} = \overline{x}$). Maximizing with respect to Σ yields $\hat{\Sigma} = \frac{1}{n}A$.

Distribution of the Quadratic Form $n(\overline{x} - \mu)'\Sigma^{-1}(\overline{x} - \mu) \sim \chi_p^2$

- Define a transformed vector $y = \sqrt{n}\Sigma^{-1/2}(\overline{x} - \mu)$.
- Find its distribution: $E(y) = \sqrt{n}\Sigma^{-1/2}(\mu - \mu) = 0$.
- Find its variance: $Var(y) = (\sqrt{n}\Sigma^{-1/2})Var(\overline{x})(\sqrt{n}\Sigma^{-1/2})' = n\Sigma^{-1/2}(\frac{1}{n}\Sigma)\Sigma^{-1/2} = I_p$.
- Thus, $y \sim N_p(0, I_p)$, which means its components y_i are iid $N(0,1)$.
- The quadratic form can be written as $y'y = \sum_{i=1}^{p} y_i^2$, which by definition is the sum of p squared independent standard normals, hence follows χ_p^2.

Conditional Distribution Independence (The Schur Complement Proof)

- To prove $x_1 - \Sigma_{12}\Sigma_{22}^{-1}x_2$ is independent of x_2, construct a transformation matrix $C = \begin{bmatrix} I_q & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I_{p-q} \end{bmatrix}$.
- Apply C to the joint vector x: $Cx = \begin{bmatrix} x_1 - \Sigma_{12}\Sigma_{22}^{-1}x_2 \\ x_2 \end{bmatrix}$.
- Calculate the new covariance matrix $C\Sigma C' = \begin{bmatrix} I_q & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I_{p-q} \end{bmatrix} \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I_q & 0 \\ -\Sigma_{22}^{-1}\Sigma_{21} & I_{p-q} \end{bmatrix}$.
- Matrix multiplication yields $\begin{bmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{bmatrix}$.
- Since the off-diagonal block is strictly 0, the transformed vector $x_1 - \Sigma_{12}\Sigma_{22}^{-1}x_2$ and x_2 are uncorrelated. For multivariate normal distributions, zero covariance implies strict independence.

Cochran's Theorem Application for Sample Variance ($\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$)

- Write the sum of squared deviations as a quadratic form: $\sum_{i=1}^{n}(Y_i - \overline{Y})^2 = y'By$, where $B = I_n - \frac{1_n 1_n'}{n}$.
- Note that B is idempotent ($B^2 = B$) and its trace is $tr(B) = n - 1$.
- Because B is symmetric and idempotent with rank $n - 1$, there exists an $(n-1) \times n$ matrix C such that $CC' = I_{n-1}$ and $C'C = B$.
- Define $z = Cy \sim N_{n-1}(0, \sigma^2 I_{n-1})$. Then $z/\sigma \sim N_{n-1}(0, I_{n-1})$.
- The sum of squares becomes $y'By = y'C'Cy = z'z$.
- Therefore, $\frac{(n-1)S^2}{\sigma^2} = \frac{z'z}{\sigma^2} \sim \chi_{n-1}^2$.

How to Prove a Matrix is Positive Semidefinite ($A \geq 0$)

• Method 1 (By Definition): Show that for any non-zero $p \times 1$ vector x, the quadratic form yields $x'Ax \geq 0$.

• Method 2 (By Eigenvalues): Calculate all eigenvalues λ_i of the matrix A. If $\lambda_i \geq 0$ for all $i = 1, \ldots, p$, then $A \geq 0$.

• Method 3 (By Matrix Factorization / Gram Structure): Show that the matrix A can be expressed in the form $B'B$ (where B is any matrix).

  • Proof: Let $y = Bx$.

  • Then evaluate the quadratic form: $x'Ax = x'(B'B)x = (Bx)'(Bx) = y'y$.

  • Since $y'y = \sum_{i=1}^{k} y_i^2$ is strictly a sum of squared real numbers, it guarantees that $y'y \geq 0$. Thus, $A \geq 0$.

  • **Note**: This Method 3 is the standard procedure used to prove that the SSCP matrix $A = (X - 1_n\overline{x}')'(X - 1_n\overline{x}')$ and the sample covariance matrix $S = \frac{1}{n-1}A$ are symmetric positive semidefinite.

Exercise

(Spectral Decomposition)

- 题目: 已知 $A = \begin{bmatrix} 1 & 1 \\ 2 & -2 \\ 2 & 2 \end{bmatrix}$，求 $A'A$ 的谱分解。

$A'A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 1 \\ 1 & 9 \end{bmatrix}$. 解 $|A'A - \lambda I_2| = 0 \implies (9-\lambda)^2 - 1 = 0 \implies \lambda_1 = 8, \lambda_2 = 10$.

- 对于 $\lambda_1 = 8$, 解 $(A'A - 8I)x = 0$, 单位化得 $h_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$. $x_1 + x_2 = 0, x_1 = 1, x_2 = -1$然后再标准化
- 对于 $\lambda_2 = 10$, 解 $(A'A - 10I)x = 0$, 单位化得 $h_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.
- $A'A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$.

Multivariate Normal Distribution & Properties

- **MVN Density**: For $X \sim N_p(\mu, \Sigma)$, the joint density function is $f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\}$.
- **Linear Transformation**: If A is a constant $q \times p$ matrix and d is a constant $q \times 1$ vector, then $AX + d \sim N_q(A\mu + d, A\Sigma A')$. • **Note (Dimensions)**: The output vector $AX + d$ has dimension $q \times 1$, and its new covariance matrix $A\Sigma A'$ has dimension $q \times q$
- **Marginal Distributions**: All subsets of X strictly follow a normal distribution. If $X = [X_1', X_2']' \sim N(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix})$, then the marginal distribution for each subset is $X_i \sim N_{p_i}(\mu_i, \Sigma_{ii})$ for $i = 1, 2$.

Linear Transformation of MVN Vector

- Problem: Given $X \sim N_3(\mu, \Sigma)$ with $\mu = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$, find the distribution of $X_1 + X_2 - 2X_3$.

Express the linear combination as $AX + d$. Here, the coefficient vector is $A = \begin{bmatrix} 1 & 1 & -2 \end{bmatrix}$ and $d = 0$.
For a multivariate normal vector, $AX \sim N(A\mu, A\Sigma A')$.

Calculate New Mean ($A\mu$): $A\mu = \begin{bmatrix} 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2 + 3 - 2 = 3$.

Calculate New Variance ($A\Sigma A'$): $A\Sigma A' = \begin{bmatrix} 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = 2$.

The transformed variable reduces to a univariate normal distribution, $X_1 + X_2 - 2X_3 \sim N(3, 2)$.

Proofs for Conditional Covariance & Positive Definiteness

- **(a) Prove $\Sigma_{11\bullet2}$ is Positive Definite**: Given the full covariance matrix $\Sigma > 0$, construct a block matrix $B = \begin{bmatrix} I \\ -\Sigma_{22}^{-1}\Sigma_{21} \end{bmatrix}$.
  - Multiply to find $B'\Sigma B$: $\begin{bmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \end{bmatrix} \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I \\ -\Sigma_{22}^{-1}\Sigma_{21} \end{bmatrix} = \begin{bmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \end{bmatrix} \begin{bmatrix} \Sigma_{11\bullet2} \\ O \end{bmatrix} = \Sigma_{11\bullet2}$.
  - Positive Definite Proof: For any non-zero vector x, since B has full column rank, the vector $y = Bx \neq 0$. The quadratic form $x'\Sigma_{11\bullet2}x = x'(B'\Sigma B)x = (Bx)'\Sigma(Bx) = y'\Sigma y$. Since $\Sigma > 0$, we perfectly know $y'\Sigma y > 0$, hence $\Sigma_{11\bullet2} > 0$.
- **(b) Prove Block Diagonal Matrix $C\Sigma C'$ is Positive Definite**: Given $C = \begin{bmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \\ O & I \end{bmatrix}$.
  - Calculate $C\Sigma C'$: First, $C\Sigma = \begin{bmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & O \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} \Sigma_{11\bullet2} & O \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$.

Then, $(C\Sigma)C' = \begin{bmatrix} \Sigma_{11\bullet2} & O \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I & O \\ -\Sigma_{22}^{-1}\Sigma_{21} & I \end{bmatrix} = \begin{bmatrix} \Sigma_{11\bullet2} & O \\ O & \Sigma_{22} \end{bmatrix}$.

  - Positive Definite Proof: For any non-zero vector y, we evaluate $y'(C\Sigma C')y$. Since C is a non-singular square matrix (its determinant $|C| = 1$), the vector $x = C'y$ must be non-zero ($x \neq 0$). Therefore, $y'(C\Sigma C')y = (C'y)'\Sigma(C'y) = x'\Sigma x$. Since $\Sigma > 0$, $x'\Sigma x > 0$, strictly proving $C\Sigma C' > 0$.

Conditional Distribution via Transformation

- **Problem**: Given $Y \sim N(\mu, \Sigma)$, find $Y_2|(Y_1 - Y_3 = z)$.where: $\mu = [1, 2, 0]', \Sigma = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$.

1. **Define Transformation**: We need the joint distribution of Y_2 and $Z = Y_1 - Y_3$. Let $X = \begin{bmatrix} Y_2 \\ Y_1 - Y_3 \end{bmatrix} = AY$, where $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$.
2. **Calculate Mean** $A\mu$: $A\mu = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mu_{Y_2} \\ \mu_Z \end{bmatrix}$.

3. **Calculate Covariance** $A\Sigma A'$:

- $A\Sigma = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & -3 \end{bmatrix}$.
- $A\Sigma A' = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & -3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$.

4. **Find Conditional Distribution**: Since $\Sigma_{12} = 0$, Y_2 and Z are **independent**.

- Result: $Y_2|Z = z \sim N(\mu_{Y_2}, \Sigma_{11}) = N(2, 2)$.

Joint and Conditional Distribution

Problem: Find (a) Joint distribution of $W = [Y_1 - 2Y_2, Y_1 - 3Y_3]'$ and (b) $(Y_1 - 2Y_2)|Y_3 = y_3$. $\mu = [2\beta, \beta, \beta]'$, $\Sigma = \begin{bmatrix} 1 & 0 & \rho \\ 0 & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix}$, with $-1 < \rho < 1$

.

(a) Joint Distribution of W

1. **Linear Transformation**: $W = AY$ where $A = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix}$.

2. **Mean** $A\mu$: $\begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} 2\beta \\ \beta \\ \beta \end{bmatrix} = \begin{bmatrix} 2\beta - 2\beta \\ 2\beta - 3\beta \end{bmatrix} = \begin{bmatrix} 0 \\ -\beta \end{bmatrix}$.

3. **Covariance** $A\Sigma A'$:

- $A\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 & \rho \\ 0 & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -\rho \\ 1 - 3\rho & -3\rho & \rho - 3 \end{bmatrix}$.
- $A\Sigma A' = \begin{bmatrix} 1 & -2 & -\rho \\ 1 - 3\rho & -3\rho & \rho - 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 0 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} 5 & 1 + 3\rho \\ 1 + 3\rho & 10 - 6\rho \end{bmatrix}$.

4. **Result**: $W \sim N_2\left(\begin{bmatrix} 0 \\ -\beta \end{bmatrix}, \begin{bmatrix} 5 & 1 + 3\rho \\ 1 + 3\rho & 10 - 6\rho \end{bmatrix}\right)$.

(b) Conditional Distribution $(Y_1 - 2Y_2)|Y_3 = y_3$

1. **New Transformation**: Let $X = \begin{bmatrix} Y_1 - 2Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} Y$.

2. **Joint Parameters**:

- Mean: $A\mu = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2\beta \\ \beta \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ \beta \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$.
- Covariance: $A\Sigma A' = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \rho \\ 0 & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -\rho \\ -\rho & 1 \end{bmatrix} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$.

3. **Apply Conditional Formula**:

- $\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_3 - \mu_2) = 0 + (-\rho)(1)^{-1}(y_3 - \beta) = -\rho(y_3 - \beta)$.
- $\Sigma_{11|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = 5 - (-\rho)(1)^{-1}(-\rho) = 5 - \rho^2$.

4. **Result**: $(Y_1 - 2Y_2)|Y_3 = y_3 \sim N(-\rho(y_3 - \beta), 5 - \rho^2)$.

Ch3 Inferences on the Mean Vector

Notation (Hypothesis Testing)

- S: Sample covariance matrix, specifically with denominator $n - 1$, defined as $S = \frac{1}{n-1} \sum_{j=1}^{n} (x_j - \overline{x})(x_j - \overline{x})'$.
- d_j: The difference vector for the j-th pair in a paired comparison, defined as $x_{1j} - x_{2j}$.
- $\overline{d}$: Sample mean vector of the differences, defined as $\overline{d} = \frac{1}{n} \sum_{j=1}^{n} d_j$.
- S_d: Sample covariance matrix of the differences, defined as $S_d = \frac{1}{n-1} \sum_{j=1}^{n} (d_j - \overline{d})(d_j - \overline{d})'$.
- S_p: Pooled sample covariance matrix for two independent samples, defined as $S_p = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$.

- C: A $q \times p$ contrast matrix of constants used for testing linear combinations.
- 1_p: A $p \times 1$ vector of ones.

5. Profile Analysis (Two-Sample Extension)

Divides the testing procedure into three sequential stages for two independent groups ($n = n_1 + n_2$).

- Stage a: Parallel Profiles 平行性
- Hypothesis: $H_0^{(1)} : C\mu_1 = C\mu_2$ (where C is the $(p-1) \times p$ difference matrix).
- Test Statistic: $T^2 = \frac{n_1 n_2}{n_1+n_2}(\overline{x}_1 - \overline{x}_2)'C'(CS_pC')^{-1}C(\overline{x}_1 - \overline{x}_2)$.
- Decision Rule: Reject $H_0^{(1)}$ if $\frac{n-p}{p-1} \cdot \frac{T^2}{n-2} > F_{p-1,n-p}(\alpha)$.
- Stage b: Coincident Profiles (Requires Stage a to NOT be rejected) 重合性
- Hypothesis: $H_0^{(2)} : 1_p'\mu_1 = 1_p'\mu_2$. Tests if the two parallel lines are at the exact same height.
  - Test Statistic (T^2 form): $T^2 = \frac{n_1 n_2}{n_1+n_2}(\overline{x}_1 - \overline{x}_2)'1_p(1_p'S_p1_p)^{-1}1_p'(\overline{x}_1 - \overline{x}_2)$. Under $H_0^{(2)}$, $T^2 \sim F_{1,n-2}$.
  - Test Statistic (t form): $t_{stat} = \frac{1_p'(\overline{x}_1 - \overline{x}_2)}{\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})(1_p'S_p1_p)}}$. Under $H_0^{(2)}$, $t_{stat} \sim t_{n-2}$.
  - Decision Rule: Reject $H_0^{(2)}$ if $T^2 > F_{1,n-2}(\alpha)$, or equivalently, if $|t_{stat}| > t_{n-2}(\frac{\alpha}{2})$.
- Stage c: Level Profiles (Requires Stage b to NOT be rejected) 水平性
- Hypothesis: $H_0^{(3)} : C\mu = 0$ (all means equal to the same constant). We estimate the common mean vector using the pooled overall sample mean $\overline{x} = \frac{n_1\overline{x}_1 + n_2\overline{x}_2}{n_1+n_2}$.
- Test Statistic: $T^2 = n\overline{x}'C'(CS_pC')^{-1}C\overline{x}$.
- Decision Rule: Reject $H_0^{(3)}$ if $\frac{n-p}{p-1} \cdot \frac{T^2}{n-2} > F_{p-1,n-p}(\alpha)$.

6. Confidence Regions & Intervals

- **Confidence Region:** The p-dimensional $100(1-\alpha)\%$ confidence region for μ is an ellipsoid consisting of all μ satisfying:

$(\overline{x} - \mu)'S^{-1}(\overline{x} - \mu) \le \frac{p(n-1)F_{p,n-p}(\alpha)}{n(n-p)}$.

- Note: Geometrically, this forms an ellipsoid centered at $\overline{x}$. Its axes lengths and directions are determined by the eigenvalues and eigenvectors of S.

Notation (Bivariate Confidence Region)

- λ_1, λ_2: Eigenvalues of the sample covariance matrix S (assuming $\lambda_1 > \lambda_2$).
- v_1, v_2: Normalized eigenvectors corresponding to λ_1 and λ_2.
- a, b, c: Elements of the inverse sample covariance matrix, $S^{-1} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$.

Bivariate ($p = 2$) Confidence Ellipse

- **Quadratic Form Expansion:** The $100(1-\alpha)\%$ confidence region for $\mu = (\mu_1, \mu_2)'$ consists of all values satisfying:
  $a(\overline{x}_1 - \mu_1)^2 + 2b(\overline{x}_1 - \mu_1)(\overline{x}_2 - \mu_2) + c(\overline{x}_2 - \mu_2)^2 \le \frac{2(n-1)F_{2,n-2}(\alpha)}{n(n-2)}$
- **Geometric Properties of the Ellipse:**
  - Center: $(\overline{x}_1, \overline{x}_2)$.
  - Major Axis: Lies along the direction of eigenvector v_1. The total length of the major axis is $2\sqrt{\frac{2(n-1)F_{2,n-2}(\alpha)\lambda_1}{n(n-2)}}$.
  - Minor Axis: Lies along the direction of eigenvector v_2 (perpendicular to the major axis). The total length of the minor axis is $2\sqrt{\frac{2(n-1)F_{2,n-2}(\alpha)\lambda_2}{n(n-2)}}$.
- **Equivalence to Hypothesis Testing:**
  To test $H_0 : \mu = (\mu_{01}, \mu_{02})'$ at significance level α, simply substitute μ_{01} and μ_{02} into the quadratic form expansion above. If the computed numerical value is $\le$ the right-hand side critical value, the point lies inside the confidence ellipse, and we do not reject H_0.

- **Simultaneous Confidence Intervals:** Guarantees the joint probability that all p intervals are simultaneously true is at least $1 - \alpha$. For individual component μ_i:

$\overline{x}_i \pm \sqrt{\frac{p(n-1)F_{p,n-p}(\alpha)}{n-p}} \cdot \sqrt{\frac{s_{ii}}{n}}$.

- **Bonferroni Method:** Often gives shorter, more precise intervals than the simultaneous method by distributing the alpha risk. For individual component μ_i:

$\overline{x}_i \pm t_{n-1}(1 - \frac{\alpha}{2p}) \cdot \sqrt{\frac{s_{ii}}{n}}$.

7. Test for Equality of Covariance Matrices (Box's M Test)

- Hypothesis: $H_0 : \Sigma_1 = \Sigma_2$ versus $H_1 : \Sigma_1 \ne \Sigma_2$.
- Test Statistic: $M = (n-2)\ln|S_p| - (n_1 - 1)\ln|S_1| - (n_2 - 1)\ln|S_2|$.
- Approximation: $(1-u)M \sim \chi^2_{p(p+1)/2}$ approximately, where the scaling factor is $u = (\frac{1}{n_1-1} + \frac{1}{n_2-1} - \frac{1}{n-2}) \cdot [\frac{2p^2+3p-1}{6(p+1)}]$.
- Note: This acts as a preliminary assumption check for the two-sample T^2 test.

8. Transformations to Near Normality
• **Counts** (y): Transformed to $\sqrt{y}$.
• **Proportions** ($\hat{p}$): Transformed using the logit function, defined as $\frac{1}{2}\log(\frac{\hat{p}}{1-\hat{p}})$.
• **Correlations** (r): Transformed using Fisher's z-transformation, defined as $z_r = \frac{1}{2}\log(\frac{1+r}{1-r})$.

• Box and Cox Transformation: Used to make non-normal univariate data more "normal looking". Given by:

$x^{(\lambda)} = \frac{x^{\lambda}-1}{\lambda}$ for $\lambda \neq 0$.

$x^{(\lambda)} = \ln(x)$ for $\lambda = 0$.

• Note: For a multivariate observation, a distinct power transformation λ_k must be selected for each of the p variables by maximizing their individual log-likelihood functions.

Notation (Box and Cox Transformation)
• λ: Power transformation parameter.
• $x^{(\lambda)}$: The transformed observation.
• $l(\lambda)$: The profile log-likelihood function to be maximized.
• x_{jk}: The j-th observation on the k-th variable in a multivariate dataset ($k = 1, 2, \ldots, p$).
• λ_k: The power transformation parameter specific to the k-th variable.
• $\hat{\lambda}_k$: The optimal value that maximizes $l(\lambda_k)$ for the k-th variable.

8. Box and Cox Transformation
• **Univariate Transformation Formula:**
$x^{(\lambda)} = \frac{x^{\lambda}-1}{\lambda}$ for $\lambda \neq 0$, and $x^{(\lambda)} = \ln(x)$ for $\lambda = 0$.
• **Finding the Optimal λ (Univariate):**
Given observations $x_1, \ldots, x_n$, the appropriate power λ is the solution that maximizes the log-likelihood expression:
$l(\lambda) = -\frac{n}{2}\ln\left[\frac{1}{n}\sum_{j=1}^{n}(x_j^{(\lambda)} - \overline{x^{(\lambda)}})^2\right] + (\lambda - 1)\sum_{j=1}^{n}\ln(x_j)$
where the sample mean of the transformed data is $\overline{x^{(\lambda)}} = \frac{1}{n}\sum_{j=1}^{n}\left(\frac{x_j^{\lambda}-1}{\lambda}\right)$.
With multivariate observations, a distinct power transformation must be selected for each of the p variables independently.
• **Finding the Optimal λ_k:**
Each λ_k is selected by maximizing its own log-likelihood function:
$l(\lambda_k) = -\frac{n}{2}\ln\left[\frac{1}{n}\sum_{j=1}^{n}(x_{jk}^{(\lambda_k)} - \overline{x_k^{(\lambda_k)}})^2\right] + (\lambda_k - 1)\sum_{j=1}^{n}\ln(x_{jk})$
where $\overline{x_k^{(\lambda_k)}} = \frac{1}{n}\sum_{j=1}^{n}\left(\frac{x_{jk}^{\lambda_k}-1}{\lambda_k}\right)$.
• **Constructing the Transformed Vector:**
Let $\hat{\lambda}_1, \hat{\lambda}_2, \ldots, \hat{\lambda}_p$ be the values that individually maximize the equation above for each variable. Then the j-th transformed multivariate observation vector becomes:
$x_j^{(\hat{\lambda})} = \left[\frac{x_{j1}^{\hat{\lambda}_1}-1}{\hat{\lambda}_1}, \frac{x_{j2}^{\hat{\lambda}_2}-1}{\hat{\lambda}_2}, \ldots, \frac{x_{jp}^{\hat{\lambda}_p}-1}{\hat{\lambda}_p}\right]'$

Ch5 Principal Component Analysis (PCA)

Notation (Principal Component Analysis)
• x: $p \times 1$ random vector of original variables $(X_1, \ldots, X_p)'$.
• Σ: Population covariance matrix of x.
• S: Sample covariance matrix, defined as $S = \frac{1}{n-1}\sum_{j=1}^{n}(x_j - \overline{x})(x_j - \overline{x})'$.
• $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p \geq 0$: Ordered eigenvalues of Σ (or S for sample calculations).
• Λ: Diagonal matrix of eigenvalues, $\Lambda = diag(\lambda_1, \ldots, \lambda_p)$.
• v_i: Normalized eigenvector corresponding to λ_i.
• Y_i: The i-th principal component (PC).
• v_{ik}: The k-th element of the i-th eigenvector v_i.
• σ_{kk}: Variance of the original variable X_k.

1. Unsupervised Learning & PCA Basics
• **Unsupervised Learning:** Only the feature matrix X is available (no labels Y). Used for dimension reduction, image compression, noise reduction, and exploratory data analysis.
• **PCA Objective:** Represents a dataset using fewer variables with negligible loss of information via linear combinations of the original variables. Formulated via Maximum Variance or Minimum Reconstruction Error.
• **Computation:** Solved via Spectral Decomposition or Singular-Value Decomposition (SVD).

2. Maximum Variance Formulation
• **First PC (Y_1):** Linear combination $Y_1 = v_1'x$. We maximize $Var(Y_1) = v_1'\Sigma v_1$ subject to the constraint $v_1'v_1 = 1$. The maximum variance is λ_1, and the loadings are v_1.
• **Second PC (Y_2):** Linear combination $Y_2 = v_2'x$. We maximize $Var(Y_2) = v_2'\Sigma v_2$ subject to $v_2'v_2 = 1$ and the orthogonality constraint $v_1'v_2 = 0$ ($v_1 \perp v_2$). The maximum variance is λ_2, and the loadings are v_2.

3. Properties of Principal Components

Notation (PCA Spectral Decomposition & Proofs)

- B, Σ: A $p \times p$ positive semidefinite covariance matrix.
- $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p \geq 0$: Ordered eigenvalues of the matrix.
- $v_1, \ldots, v_p$: Normalized eigenvectors corresponding to the eigenvalues.
- P: Orthogonal matrix composed of eigenvectors, defined as $P = [v_1, \ldots, v_p]$, where $P'P = I$.
- Λ: Diagonal matrix of eigenvalues, defined as $\Lambda = diag(\lambda_1, \ldots, \lambda_p)$.
- l: Any non-zero $p \times 1$ vector representing linear combination weights.
- u: A transformed vector defined as $u = P'l$.
- e_i: A $p \times 1$ indicator vector of 0s except that the i-th element is 1.

Lemma 1: Rayleigh Quotient Maximization

Suppose the $p \times p$ positive semidefinite matrix B has ordered eigenvalues $\lambda_1 \geq \ldots \geq \lambda_p \geq 0$ and corresponding normalized eigenvectors $v_1, \ldots, v_p$. Then:

- i) $\max_{l \neq 0} \frac{l'Bl}{l'l} = \lambda_1$, and the maximum is attained at $l = v_1$.
- ii) $\min_{l \neq 0} \frac{l'Bl}{l'l} = \lambda_p$, and the minimum is attained at $l = v_p$.
- iii) $\max_{l \perp v_1, \ldots, v_k} \frac{l'Bl}{l'l} = \lambda_{k+1}$, and the maximum is attained at $l = v_{k+1}$ for $k = 1, 2, \ldots, p-1$.

Proof of Lemma 1 (Maximization)

- Let $B = P\Lambda P'$ be the spectral decomposition of B. Let $u = P'l$.
- The objective function becomes: $\frac{l'Bl}{l'l} = \frac{l'P\Lambda P'l}{l'PP'l} = \frac{u'\Lambda u}{u'u}$.
- Expanding into summations: $\frac{u'\Lambda u}{u'u} = \frac{\sum_{i=1}^{p} \lambda_i u_i^2}{\sum_{i=1}^{p} u_i^2}$.
- Since λ_1 is the largest eigenvalue, $\frac{\sum_{i=1}^{p} \lambda_i u_i^2}{\sum_{i=1}^{p} u_i^2} \leq \lambda_1 \frac{\sum_{i=1}^{p} u_i^2}{\sum_{i=1}^{p} u_i^2} = \lambda_1$.
- When $l = v_1$, $P'v_1 = (1, 0, \ldots, 0)'$, which yields exactly λ_1.
- Note: For part (iii), the constraint $l \perp v_1, \ldots, v_k$ implies $v_i'l = 0$ for $i \leq k$. Thus $u_i = 0$ for $i \leq k$, meaning the summation starts from $k+1$, shifting the upper bound to λ_{k+1}.

Proof: Uncorrelated PCs and Variance

Using the spectral decomposition $\Sigma = P\Lambda P'$, all PCs are strictly uncorrelated and their variance equals their respective eigenvalue.

- **Zero Covariance:** $Cov(Y_i, Y_j) = Cov(v_i'x, v_j'x) = v_i'\Sigma v_j = v_i'P\Lambda P'v_j = e_i'\Lambda e_j = 0$ (for $i \neq j$).
- **PC Variance:** $Var(Y_i) = v_i'\Sigma v_i = v_i'P\Lambda P'v_i = e_i'\Lambda e_i = \lambda_i$.

Proposition 2: Total Variation Conservation

The total variation of the original variables $X_1, \ldots, X_p$ is exactly equal to the total variation of the transformed PC variables $Y_1, \ldots, Y_p$.

- Equation: $\sum_{i=1}^{p} Var(X_i) = \sum_{i=1}^{p} Var(Y_i)$.

Proof of Proposition 2 (The Trace Trick)

- The sum of variances of the original variables is the trace of the covariance matrix: $\sum_{i=1}^{p} Var(X_i) = tr(\Sigma)$.
- Substitute the spectral decomposition: $tr(\Sigma) = tr(P\Lambda P')$.
- Apply the cyclic property of the trace operation $tr(AB) = tr(BA)$: $tr(P\Lambda P') = tr(\Lambda P'P)$.
- Since P is orthogonal ($P'P = I$), we have $tr(\Lambda P'P) = tr(\Lambda)$.

- Since Λ is a diagonal matrix containing all eigenvalues, $tr(\Lambda) = \sum_{i=1}^{p} \lambda_i = \sum_{i=1}^{p} Var(Y_i)$.

- **Uncorrelated PCs:** $Cov(Y_i, Y_j) = 0$ for $i \neq j$.
- **Variance:** $Var(Y_i) = \lambda_i$.
- **Total Variation:** The total variation of the original variables equals the total variation of the PCs: $\sum_{i=1}^{p} Var(X_i) = \sum_{i=1}^{p} Var(Y_i) = tr(\Sigma) = \sum_{i=1}^{p} \lambda_i$.
- **Proportion of Variance Explained:** The importance of Y_i is $\frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j}$. Retaining k PCs explains $\frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{p} \lambda_j}$ of the total variance.

- **Covariance between X and Y:** $Cov(X_k, Y_i) = v_{ik}\lambda_i$.
- **Correlation between X and Y:** $Corr(X_k, Y_i) = \frac{v_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}$.

4. Large Sample Properties & Inference

Assume $x_1, \ldots, x_n \sim N_p(\mu, \Sigma)$ with distinct positive eigenvalues.
- **Eigenvalue Distribution:** $\sqrt{n}(\hat{\lambda} - \lambda) \sim N_p(0, 2\Lambda^2)$ asymptotically. Thus, $\hat{\lambda}_i$ is approximately $N(\lambda_i, \frac{2\lambda_i^2}{n})$.
- **Eigenvector Distribution:** $\sqrt{n}(\hat{v}_i - v_i) \sim N_p(0, A_i)$ where $A_i = \lambda_i \sum_{k \neq i} \frac{\lambda_k}{(\lambda_k - \lambda_i)^2} v_k v_k'$.
- **Independence:** Each $\hat{\lambda}_i$ is distributed independently of the elements of its associated $\hat{v}_i$.
- **Confidence Interval for λ_i:** A large sample $100(1-v)\%$ CI is $\frac{\hat{\lambda}_i}{1+z_{v/2}\sqrt{\frac{2}{n}}} \leq \lambda_i \leq \frac{\hat{\lambda}_i}{1-z_{v/2}\sqrt{\frac{2}{n}}}$.

5. Practical Considerations & Extensions

- **Standardization:** If units of measurement are artificial or vastly different, standardize variables before PCA by using the Correlation Matrix where $tr(R) = p$. If units carry physical meaning or information (e.g., gene expression), do NOT standardize to unit variance. $Z_i = \frac{X_i - \mu_i}{\sigma_i}$ for $i = 1, \ldots, p$.
- **Scree Plot:** A line plot of the variances (eigenvalues) against the order of the PC. Used to visually determine the suitable number of PCs to retain.
- **Loadings Interpretation:** Loadings v_i are unique only up to a sign change. They represent the relative importance of original variables in forming the PC (e.g., contrasting speed vs. power).
- **Extensions:**
  - a) Probabilistic PCA / Exponential family PCA (for explicitly modeling binary/count data).
  - b) Nonlinear dimension reduction: Kernel PCA, Variational Autoencoder (VAE, a non-linear method based on neural networks).

1. Singular-Value Decomposition (SVD)

- **Definition:** Any $n \times p$ mean-centered matrix X can be decomposed as $X = U\Sigma V'$.

- **Relationship to PCA:**

  i) By expansion, $X'X = V\Sigma^2 V'$. Therefore, the columns of V are the exact eigenvectors of $X'X$, which act as the principal component loadings v_i.

  ii) The singular values σ_i are linked to the eigenvalues λ_i of the sample covariance matrix S by: $\lambda_i = \frac{\sigma_i^2}{n-1}$.

  iii) The principal component scores are easily computed without V via $XV = U\Sigma V'V = U\Sigma$.

- **Advantage:** SVD is more numerically stable and computationally efficient than calculating $X'X$ explicitly for spectral decomposition, avoiding rounding errors in ill-conditioned matrices.

2. Minimum Reconstruction Error Formulation

- **Objective:** PCA seeks to find a k-dimensional subspace such that the squared Euclidean distance between the original data points x_j and their projected approximations $\hat{x}_j$ onto this subspace is minimized.

- **Optimization Problem:** $\min_{V_k} \sum_{j=1}^{n} ||x_j - V_k V_k' x_j||^2$

  where V_k consists of the first k orthonormal columns of the loadings matrix V.

- **Reconstruction Formula:** The best k-rank approximation of an original uncentered observation x_j adds the projection back to the sample mean: $\hat{x}_j = \overline{x} + \sum_{i=1}^{k} (v_i'(x_j - \overline{x}))v_i$

- **Total Minimum Error:** The minimum reconstruction error (residual sum of squares) is exactly equal to the sum of the discarded eigenvalues adjusted by the sample size denominator: $Error = (n-1)\sum_{i=k+1}^{p} \lambda_i$

- **Conclusion:** Maximizing the captured variance (which retains the largest λ_i) is mathematically identical to minimizing the reconstruction error (which discards the smallest λ_i).

Ch6 Partial Correlation Analysis

Notation (Partial Correlation Analysis)

- X, Y, X_i, X_j: Univariate random variables.

- z: A $q \times 1$ random vector representing the control variables.
- ρ_{XY}: Simple population correlation coefficient.
- r_{XY}: Simple sample correlation coefficient.
- $\Sigma_{11\bullet z}$: Conditional population covariance matrix of the remaining variables given z.
- $S_{11\bullet z}$: Conditional sample covariance matrix given z.
- $\sigma_{ij\bullet z}$: The (i, j)-th element of $\Sigma_{11\bullet z}$.
- $s_{ij\bullet z}$: The (i, j)-th element of $S_{11\bullet z}$.
- $\rho_{ij\bullet z}$: Population partial correlation coefficient between X_i and X_j given z.
- $r_{ij\bullet z}$: Sample partial correlation coefficient between X_i and X_j given z.

1. Simple Correlation Coefficient

- **Sample Estimator:** $r_{XY} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{(\sum_{i=1}^{n}(x_i-\overline{x})^2)(\sum_{i=1}^{n}(y_i-\overline{y})^2)}}$
- **Hypothesis Testing:**

  To test $H_0 : \rho_{XY} = 0$ versus $H_1 : \rho_{XY} \neq 0$:

Test Statistic: $t = \sqrt{n-2} \cdot \frac{r_{XY}}{\sqrt{1-r_{XY}^2}} \sim t_{n-2}$. Decision Rule: Reject H_0 if $|t| > t_{n-2}(\frac{\alpha}{2})$.

- **Limitation:** A large simple correlation does not imply causation, as it may be confounded by a third common variable z affecting both X and Y.

2. Partial Correlation Coefficient

Used to measure the linear association between X_i and X_j while controlling for the effects of a third party z (a single variable or a group of q variables).

- **Population Conditional Covariance:** Based on the partitioned covariance matrix: $\Sigma_{11\bullet z} = \Sigma_{11} - \Sigma_{1z}\Sigma_{zz}^{-1}\Sigma_{z1}$.
- **Population Partial Correlation:** $\rho_{ij\bullet z} = \frac{\sigma_{ij\bullet z}}{\sqrt{\sigma_{ii\bullet z}}\sqrt{\sigma_{jj\bullet z}}}$.
- **Sample Conditional Covariance:** $S_{11\bullet z} = S_{11} - S_{1z}S_{zz}^{-1}S_{z1}$.
- **Sample Partial Correlation:** $r_{ij\bullet z} = \frac{s_{ij\bullet z}}{\sqrt{s_{ii\bullet z}}\sqrt{s_{jj\bullet z}}}$.

3. Hypothesis Testing for Partial Correlation

To test whether the partial correlation is significantly different from zero, accounting for the q variables being controlled:

- **Hypothesis:** $H_0 : \rho_{ij\bullet z} = 0$ versus $H_1 : \rho_{ij\bullet z} \neq 0$.
- **Test Statistic:** $t = \sqrt{n-q-2} \cdot \frac{r_{ij\bullet z}}{\sqrt{1-r_{ij\bullet z}^2}} \sim t_{n-q-2}$.
- **Decision Rule:** Reject H_0 if $|t| > t_{n-q-2}(\frac{\alpha}{2})$.
  - Note: The degrees of freedom drop by q (the number of variables controlled for) compared to the simple correlation test.

4. Special Case ($p = 2, q = 1$)

When controlling for exactly one variable (e.g., $z = X_3$) to find the partial correlation between X_1 and X_2, matrix inversion is not required.

- **Population Formula:** $\rho_{12\bullet 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1-\rho_{13}^2)(1-\rho_{23}^2)}}$.
- **Sample Formula:** $r_{12\bullet 3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}$.

• Derivation of the formula for $p = 2, q = 1$ (Controlling for a single variable X_3):

• Let the correlation matrix of $\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ be $R = \begin{bmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{bmatrix}$.

• Here, $R_{11} = \begin{bmatrix} 1 & \rho_{12} \\ \rho_{12} & 1 \end{bmatrix}$, $R_{1z} = \begin{bmatrix} \rho_{13} \\ \rho_{23} \end{bmatrix}$, $R_{z1} = [\rho_{13} \quad \rho_{23}]$, and $R_{zz} = [1]$.

• By the Schur complement formula, the conditional covariance matrix is: $R_{11\bullet 3} = R_{11} - R_{1z}R_{zz}^{-1}R_{z1}$.

• Expand the term: $R_{1z}R_{zz}^{-1}R_{z1} = \begin{bmatrix} \rho_{13} \\ \rho_{23} \end{bmatrix} [1]^{-1} [\rho_{13} \quad \rho_{23}] = \begin{bmatrix} \rho_{13}^2 & \rho_{13}\rho_{23} \\ \rho_{13}\rho_{23} & \rho_{23}^2 \end{bmatrix}$.

• Subtract from R_{11}: $R_{11\bullet 3} = \begin{bmatrix} 1 - \rho_{13}^2 & \rho_{12} - \rho_{13}\rho_{23} \\ \rho_{12} - \rho_{13}\rho_{23} & 1 - \rho_{23}^2 \end{bmatrix}$.

• The partial correlation is the off-diagonal element divided by the square root of the product of the diagonal elements:

• $\rho_{12\bullet 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1-\rho_{13}^2)(1-\rho_{23}^2)}}$.

• Conceptual Insight: Simpson's Paradox (The Iris Dataset Case)

• Problem: The simple correlation between Sepal Length and Sepal Width is negative (-0.1176). However, the partial correlation controlling for Petal Length and Petal Width is positive (0.6286). What is going on?

• Explanation: This is a classic example of Simpson's Paradox. Different species of Iris have drastically different overall sizes. When we look at the data *as a whole* (simple correlation), a species with a very long but narrow sepal is mixed with a species with a short but wide sepal, creating an artificial negative linear trend.

• By controlling for Petal Length and Width (which act as strong proxies for the overall scale/species of the flower), we are effectively looking at the correlation *within* a specific plant size constraint. Once the confounding size factor is removed, the true biological relationship emerges: flowers with longer sepals naturally tend to have wider sepals (+0.6286).

Ch1 Matrix Operations & Important Proofs (Exam Additions)

1. Advanced Matrix Properties & Proofs

• **Proof: Eigenvalues of a Positive Definite Matrix are strictly positive ($A > 0 \implies \lambda_i > 0$)**

Let $A > 0$. By definition, $x'Ax > 0$ for any non-zero vector x. Let λ be an eigenvalue of A with a corresponding normalized eigenvector v (so $v'v = 1$ and $v \neq 0$). By definition, $Av = \lambda v$. Pre-multiplying both sides by v', we get $v'Av = v'(\lambda v) = \lambda(v'v) = \lambda$. Since $A > 0$ and $v \neq 0$, the quadratic form $v'Av > 0$. Therefore, $\lambda > 0$.

• **Proof: Non-zero eigenvalues of AB and BA are the same**

Let $\lambda \neq 0$ be an eigenvalue of AB with corresponding eigenvector $v \neq 0$. Thus, $(AB)v = \lambda v$. Pre-multiply both sides by B: $B(AB)v = B(\lambda v)$, which groups to $(BA)(Bv) = \lambda(Bv)$. Let $u = Bv$. If $u = 0$, then $(AB)v = A(0) = 0$, which would mean $\lambda v = 0$. Since $v \neq 0$, this would force $\lambda = 0$, contradicting our assumption that $\lambda \neq 0$. Thus, $u \neq 0$. This means u is a valid eigenvector for BA corresponding to the exact same eigenvalue λ.

• **Proof: The Matrix Identity** $(A + B)^{-1} = A^{-1} - A^{-1}(A^{-1} + B^{-1})^{-1}A^{-1}$

To prove this, multiply the right-hand side by $(A + B)$ and show it equals the identity matrix I:

$[A^{-1} - A^{-1}(A^{-1} + B^{-1})^{-1}A^{-1}](A + B)$

$= A^{-1}(A + B) - A^{-1}(A^{-1} + B^{-1})^{-1}A^{-1}(A + B)$

$= (I + A^{-1}B) - A^{-1}(A^{-1} + B^{-1})^{-1}(I + A^{-1}B)$

Factor out $(I + A^{-1}B)$ to the right:

$= [I - A^{-1}(A^{-1} + B^{-1})^{-1}](I + A^{-1}B)$

Notice that $(I + A^{-1}B) = A^{-1}(A + B) = A^{-1}(BB^{-1}A + BA^{-1}A)\ldots$ A simpler algebraic path from step 2 is to expand the $A^{-1}(A + B)$ explicitly:

$= I + A^{-1}B - A^{-1}(A^{-1} + B^{-1})^{-1}(A^{-1}A + A^{-1}B)$

$= I + A^{-1}B - A^{-1}(A^{-1} + B^{-1})^{-1}(A^{-1} + B^{-1})B$

Since $(A^{-1} + B^{-1})^{-1}(A^{-1} + B^{-1}) = I$, the expression simplifies to:

$= I + A^{-1}B - A^{-1}IB = I + A^{-1}B - A^{-1}B = I$. The identity holds.

Ch2 MVN Distributions (Exam Additions)

1. Constructing Joint Distribution from Marginal and Conditional

• **Problem:** If $X_1 \sim N_r(\mu_1, \Sigma_{11})$ and $(X_2|X_1 = x_1) \sim N_{p-r}(Ax_1 + b, \Omega)$, prove that the joint vector $X = [X_1', X_2']'$ is multivariate normal and find its parameters.

• **Mean Vector:** $E(X_2) = E(E(X_2|X_1)) = E(AX_1 + b) = A\mu_1 + b$. Thus $\mu = \begin{bmatrix} \mu_1 \\ A\mu_1 + b \end{bmatrix}$.

• **Cross-Covariance:** We can write $X_2 = AX_1 + b + \varepsilon$, where $\varepsilon \sim N(0, \Omega)$ is strictly independent of X_1. Then $Cov(X_1, X_2) = Cov(X_1, AX_1 + b + \varepsilon) = Cov(X_1, AX_1) + Cov(X_1, \varepsilon) = \Sigma_{11}A' + 0 = \Sigma_{11}A'$.

• **Variance of X_2:** $Var(X_2) = Var(E(X_2|X_1)) + E(Var(X_2|X_1)) = Var(AX_1 + b) + E(\Omega) = A\Sigma_{11}A' + \Omega$.

• **Joint Distribution:** $X \sim N_p\left(\begin{bmatrix} \mu_1 \\ A\mu_1 + b \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{11}A' \\ A\Sigma_{11} & \Omega + A\Sigma_{11}A' \end{bmatrix}\right)$.

2. Independence of Linear Combinations

- **Method:** To find coefficients α, β such that a linear combination $Y = \beta X_1 + \alpha X_2 - X_3$ is independent of X_3, you must set their covariance to zero.
- **Calculation:** $Cov(Y, X_3) = \beta Cov(X_1, X_3) + \alpha Cov(X_2, X_3) - Var(X_3) = 0$. Extract the known variances/covariances from Σ and solve the linear equation for α and β.

Ch3 Inferences on the Mean Vector (Exam Additions)

1. Student's t-Distribution from Sample Covariance (Proof)

- **Theorem:** Given $x_1, \ldots, x_n \sim iid\ N_p(\mu, \Sigma)$ and any arbitrary non-zero constant vector a, prove that $\frac{a'\overline{x} - a'\mu}{\sqrt{\frac{1}{n}a'Sa}} \sim t_{n-1}$.
- **Proof:**
  1. Define a univariate transformation $y_i = a'x_i$. Because linear combinations of MVN are normal, $y_i \sim N(a'\mu, a'\Sigma a)$.
  2. The sample mean of y is $\overline{y} = a'\overline{x}$, and its true theoretical variance is $\frac{1}{n}a'\Sigma a$. Standardizing this yields $Z = \frac{a'\overline{x} - a'\mu}{\sqrt{\frac{1}{n}a'\Sigma a}} \sim N(0, 1)$.
  3. The sample variance of y is $s_y^2 = \frac{1}{n-1}\sum_{i=1}^n (a'x_i - a'\overline{x})^2 = a'Sa$, where $S = \frac{1}{n-1}\sum_{i=1}^n (x_i - \overline{x})(x_i - \overline{x})'$.
  4. By Cochran's Theorem for univariate normals, $V = \frac{(n-1)s_y^2}{Var(y_i)} = \frac{(n-1)a'Sa}{a'\Sigma a} \sim \chi^2_{n-1}$.
  5. Because $\overline{x}$ and S are strictly independent, Z and V are independent.
  6. By the definition of the t-distribution, $T = \frac{Z}{\sqrt{V/(n-1)}}$. Substituting Z and V:

$$T = \frac{\frac{a'\overline{x} - a'\mu}{\sqrt{\frac{1}{n}a'\Sigma a}}}{\sqrt{\frac{a'Sa}{a'\Sigma a}}} = \frac{a'\overline{x} - a'\mu}{\sqrt{\frac{1}{n}a'Sa}} \sim t_{n-1}$$

Ch5 Principal Component Analysis (Exam Additions)

1. Rayleigh Quotient Proof for Minimum Variance (λ_p)

- **Theorem:** $min_{l\neq 0}\frac{l'\Sigma l}{l'l} = \lambda_p$, and the minimum is attained at $l = v_p$.
- **Proof:** Use the spectral decomposition $\Sigma = P\Lambda P'$. Let $u = P'l$. Since P is orthogonal, $l'l = u'PP'u = u'u$. Substituting this gives the quotient $\frac{u'\Lambda u}{u'u} = \frac{\sum_{i=1}^p \lambda_i u_i^2}{\sum_{i=1}^p u_i^2}$. Because λ_p is the smallest eigenvalue, $\lambda_i \geq \lambda_p$ for all i. Therefore, $\sum \lambda_i u_i^2 \geq \lambda_p \sum u_i^2$. The quotient is strictly $\geq \lambda_p$. The absolute minimum λ_p is attained when $u_p = 1$ and all other $u_i = 0$, which means $u = [0 \quad \ldots \quad 0 \quad 1]'$. Since $l = Pu$, this perfectly corresponds to $l = v_p$.

2. Inverse Sample Covariance (S^{-1}) & PCA

- **Proof that $S^{-1} = H\Lambda^{-1}H'$:**
  Given the spectral decomposition $S = H\Lambda H'$, where H is orthogonal ($H'H = I$). We multiply S by our proposed inverse:
  $S(H\Lambda^{-1}H') = (H\Lambda H')(H\Lambda^{-1}H') = H\Lambda(H'H)\Lambda^{-1}H'$.
  Since $H'H = I$, this simplifies to $H\Lambda I\Lambda^{-1}H' = H(\Lambda\Lambda^{-1})H' = HIH' = HH' = I$.
- **Minimizing the Inverse Rayleigh Quotient:**
  To find a vector u that minimizes $\frac{u'S^{-1}u}{u'u}$, we must find the smallest eigenvalue of S^{-1}. The eigenvalues of S^{-1} are exactly $1/\lambda_i$. Since $\lambda_1 > \lambda_2 > \cdots > \lambda_p$, the smallest eigenvalue of S^{-1} is $1/\lambda_1$. According to the Rayleigh quotient theorem, this minimum value is achieved when the vector u is the eigenvector corresponding to $1/\lambda_1$, which is simply v_1 (the loadings of the first principal component).

Question 2: Principal Component Analysis

- **Background:** Given 3-dimensional data with sample mean $\overline{x} = [1, 3, 2]'$. The eigenvalues of the covariance matrix S are $\lambda_1 = 18.77$, $\lambda_2 = 6.41$, $\lambda_3 = 0.02$, with corresponding eigenvectors h_1, h_2, h_3.

(a) Find the percentage of total variance explained by the first three principal components

- Total variance $= \text{tr}(S) = 18.77 + 6.41 + 0.02 = 25.20$.
- PC 1 percentage: $\frac{18.77}{25.20} = 74.48\%$.
- PC 2 percentage: $\frac{6.41}{25.20} = 25.44\%$.
- PC 3 percentage: $\frac{0.02}{25.20} = 0.08\%$.

(b) Given an observation $x_i = [-1, 2, 2]'$, find its first and second principal component scores

- Centered observation: $x_c = x_i - \overline{x} = [-2, -1, 0]'$.
- First PC score: $y_1 = h_1'x_c = (0.711)(-2) + (-0.446)(-1) + (-0.544)(0) = -0.976$.
- Second PC score: $y_2 = h_2'x_c = (-0.011)(-2) + (-0.766)(-1) + (0.643)(0) = 0.788$.

(c) Find the reconstructed $\tilde{x}_i$ using only the first two principal components

- Reconstruction formula: $\tilde{x}_i = \overline{x} + y_1 h_1 + y_2 h_2$.
- $\tilde{x}_i = \begin{bmatrix}1\\3\\2\end{bmatrix} + (-0.976)\begin{bmatrix}0.711\\-0.446\\-0.544\end{bmatrix} + (0.788)\begin{bmatrix}-0.011\\-0.766\\0.643\end{bmatrix} = \begin{bmatrix}0.2974\\2.8317\\3.0377\end{bmatrix}$.

(d) Find a vector u to maximize $\frac{u'S^{-1}u}{u'u}$ and state the maximum value. Is it unique?

- Maximum value: The largest eigenvalue of S^{-1}, which is $\frac{1}{\lambda_3} = \frac{1}{0.02} = 50$.
- Corresponding vector u: The eigenvector $h_3 = [-0.703, -0.463, -0.540]'$.
- Uniqueness: Not unique. Any non-zero scalar multiple (e.g. $-h_3$) yields the same maximum.

(e) Let $Y = [2X_1, 3X_2, 0.1X_3]'$. Compute the sample covariance and correlation matrix for Y

- Let $C = \text{diag}(2, 3, 0.1)$. Then $S_Y = CS_XC'$.
- $S_Y = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} 9.5 & -6 & -7.2 \\ -6 & 7.5 & 1.4 \\ -7.2 & 1.4 & 8.2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} = \begin{bmatrix} 38.0 & -36.0 & -1.44 \\ -36.0 & 67.5 & 0.42 \\ -1.44 & 0.42 & 0.082 \end{bmatrix}$.
- Scaling does not change correlation: $R_Y = R_X$.
- $r_{12} = \frac{-6}{\sqrt{9.5 \cdot 7.5}} = -0.7108$, $r_{13} = \frac{-7.2}{\sqrt{9.5 \cdot 8.2}} = -0.8162$, $r_{23} = \frac{1.4}{\sqrt{7.5 \cdot 8.2}} = 0.1785$.
- $R_Y = \begin{bmatrix} 1 & -0.7108 & -0.8162 \\ -0.7108 & 1 & 0.1785 \\ -0.8162 & 0.1785 & 1 \end{bmatrix}$.

Question 3: Partial Correlation Analysis

- **Background:** Given $n = 100$ observations of 3 variables, with sample covariance matrix elements: $s_{11} = 3.61$, $s_{22} = 6.25$, $s_{33} = 1.21$, $s_{12} = 3.8$, $s_{13} = -0.209$, $s_{23} = -0.55$.

(a) Compute the partial correlation coefficient $r_{23\bullet 1}$

- Conditional covariance: $s_{ij\bullet 1} = s_{ij} - \frac{s_{i1} \cdot s_{j1}}{s_{11}}$.
- $s_{22\bullet 1} = 6.25 - \frac{3.8 \cdot 3.8}{3.61} = 2.25$.
- $s_{33\bullet 1} = 1.21 - \frac{(-0.209) \cdot (-0.209)}{3.61} = 1.1979$.
- $s_{23\bullet 1} = -0.55 - \frac{3.8 \cdot (-0.209)}{3.61} = -0.33$.
- Partial correlation: $r_{23\bullet 1} = \frac{s_{23\bullet 1}}{\sqrt{s_{22\bullet 1} \cdot s_{33\bullet 1}}} = \frac{-0.33}{\sqrt{2.25 \cdot 1.1979}} = -0.2010$.

(b) Test $H_0 : \rho_{23\bullet 1} = 0$ at 5% significance level

- Test Statistic: $t_{stat} = \sqrt{n - q - 2} \cdot \frac{r_{23\bullet 1}}{\sqrt{1 - r_{23\bullet 1}^2}}$, where $n = 100, q = 1$.
- $t_{stat} = \sqrt{97} \cdot \frac{-0.2010}{\sqrt{1 - (-0.2010)^2}} = -2.0208$.
- Critical Value: $t_{97}(0.025) = 1.9847$.
- Conclusion: Since $|-2.0208| > 1.9847$, reject H_0.

1. Geometric View and Reconstruction

Problem: Let $p = 2$ and suppose we have 4 centered observations ($\overline{x} = 0$): $x_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $x_4 = \begin{bmatrix} -4 \\ -1 \end{bmatrix}$. The first eigenvector (first PC direction) is $v_1 = (1, 0)^T$.

a) Compute the projection score $d_{i1} = x_i^T v_1$ for each observation.
b) Compute the reconstructed points $\tilde{x}_i = v_1 v_1^T x_i$ using only the first PC.
c) Compute the reconstruction error $\|x_i - \tilde{x}_i\|^2$ for each observation.
d) Verify that $d_{i1}^2 + \|x_i - \tilde{x}_i\|^2 = \|x_i\|^2$ for each i.

- a) $d_{11} = \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3 \quad d_{21} = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -1 \quad d_{31} = \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \quad d_{41} = \begin{bmatrix} -4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -4$
- b) $\tilde{x}_1 = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, $\tilde{x}_2 = -1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $\tilde{x}_3 = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\tilde{x}_4 = -4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}$.
- c) For $i = 1$: $\| \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \end{bmatrix} \|^2 = \| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \|^2 = 1$ For $i = 2$: $\| \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} \|^2 = \| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \|^2 = 1$
  For $i = 3$: $\| \begin{bmatrix} 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \|^2 = \| \begin{bmatrix} 0 \\ -1 \end{bmatrix} \|^2 = 1$ For $i = 4$: $\| \begin{bmatrix} -4 \\ -1 \end{bmatrix} - \begin{bmatrix} -4 \\ 0 \end{bmatrix} \|^2 = \| \begin{bmatrix} 0 \\ -1 \end{bmatrix} \|^2 = 1$
- d) For $i = 1$: $3^2 + 1 = 10$. Original norm squared is $3^2 + 1^2 = 10$. Valid.
  For $i = 2$: $(-1)^2 + 1 = 2$. Original norm squared is $(-1)^2 + 1^2 = 2$. Valid.
  For $i = 3$: $2^2 + 1 = 5$. Original norm squared is $2^2 + (-1)^2 = 5$. Valid.
  For $i = 4$: $(-4)^2 + 1 = 17$. Original norm squared is $(-4)^2 + (-1)^2 = 17$. Valid.

2. Computation — Non-Centered Data

Problem: Let $p = 2$ and consider three (non-centered) observations: $x_1 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, $x_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$, $x_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. The first eigenvector of the sample covariance is $v_1 = \frac{1}{\sqrt{2}}(1, 1)^T$.

a) Compute the sample mean $\overline{x}$.
b) Compute the reconstruction of each x_i using only 1 PC: $\tilde{x}_i = v_1 v_1^T (x_i - \overline{x}) + \overline{x}$.
c) Give a geometric interpretation of this formula.

Solution:

- a) $\overline{x} = \frac{1}{3}(\begin{bmatrix} 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 6 \\ 7 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix}) = \frac{1}{3} \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.
- b) The projection matrix is $v_1 v_1^T = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$.
  For $i = 1$: $x_1 - \overline{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. Projection is $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. Thus $\tilde{x}_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$.
  For $i = 2$: $x_2 - \overline{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Projection is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Thus $\tilde{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$.
  For $i = 3$: $x_3 - \overline{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Projection is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Thus $\tilde{x}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.
- c) Geometric interpretation: The formula shifts the origin to the sample mean $\overline{x}$, orthogonally projects the centered data onto the 1-dimensional subspace spanned by v_1, and then shifts the coordinates back to the original space. It finds the closest point on the line passing through $\overline{x}$ with

direction v_1.

3. Proof — SVD and PCA Reconstruction Connection

Problem: Let X be an $n \times p$ data matrix ($n > p$) with sample mean $\overline{x}$ and let $Y = \frac{1}{\sqrt{n-1}}(X - 1_n\overline{x}^T)$ with SVD $Y = UDV^T$. Show that the rank-q PCA reconstruction of the data can be written as: $\tilde{X} = 1_n\overline{x}^T + \sqrt{n-1}\sum_{j=1}^{q} d_j u_j v_j^T$.

Solution:

• From the SVD of the scaled and centered matrix $Y = UDV^T$, we can express it as a sum of rank-1 matrices: $Y = \sum_{j=1}^{p} d_j u_j v_j^T$.
• By the Eckart-Young-Mirsky theorem, the best rank-q approximation of Y is obtained by truncating the series to the first q components: $Y_q = \sum_{j=1}^{q} d_j u_j v_j^T$.
• The PCA reconstruction aims to approximate the original uncentered matrix X using this rank-q representation. From the definition $Y = \frac{1}{\sqrt{n-1}}(X - 1_n\overline{x}^T)$, we replace Y with its approximation Y_q and X with the reconstructed matrix $\tilde{X}$:
$Y_q = \frac{1}{\sqrt{n-1}}(\tilde{X} - 1_n\overline{x}^T)$
• Multiplying both sides by $\sqrt{n-1}$ yields: $\tilde{X} - 1_n\overline{x}^T = \sqrt{n-1}Y_q$
• Substitute Y_q back into the equation: $\tilde{X} - 1_n\overline{x}^T = \sqrt{n-1}\sum_{j=1}^{q} d_j u_j v_j^T$
• Finally, isolate $\tilde{X}$ by adding the mean matrix to both sides: $\tilde{X} = 1_n\overline{x}^T + \sqrt{n-1}\sum_{j=1}^{q} d_j u_j v_j^T$.

Original Concept: Marginal vs. Joint Gaussian

• **Question:** Provide an example where each variable in a random vector marginally follows a univariate Gaussian distribution, but jointly the random vector does not follow a multivariate Gaussian distribution.
• **Answer & Construction:**
Let $X \sim N(0, 1)$ be a standard normal random variable.
Let Z be a discrete random variable independent of X, such that $P(Z = 1) = 0.5$ and $P(Z = -1) = 0.5$.
Define $Y = Z \cdot X$.
• 1. The marginal distribution of X is explicitly $N(0, 1)$.
• 2. The marginal distribution of Y is also $N(0, 1)$ due to the perfect symmetry of the normal distribution around zero.
• 3. However, the joint distribution of $(X, Y)'$ is not bivariate normal. The probability $P(|X| = |Y|) = 1$, which implies the entire probability mass is concentrated strictly on the intersecting lines $Y = X$ and $Y = -X$. A true non-degenerate bivariate normal distribution must form a continuous 2D bell-shaped surface, not an "X-shaped" degenerate support.

Variation 1: Uncorrelated but Dependent

• **Question:** For multivariate normal distributions, being uncorrelated implies independence. Provide an example of two normally distributed variables that are uncorrelated but are NOT independent.
• **Answer & Construction:** 同上

Variation 2: Non-Gaussian Linear Combination

• **Question:** If a random vector is multivariate normal, any linear combination of its components must be univariate normal. Provide an example where X and Y are marginally normal, but their sum $X + Y$ is NOT normal.
• **Answer & Construction:**
Using the same construction: $X \sim N(0, 1)$, $P(Z = 1) = P(Z = -1) = 0.5$, and $Y = Z \cdot X$.
Consider the linear combination $W = X + Y = X + Z \cdot X = X \cdot (1 + Z)$.
• If $Z = -1$ (which occurs with 50% probability), $W = X \cdot (1 - 1) = 0$.
• If $Z = 1$ (which occurs with 50% probability), $W = X \cdot (1 + 1) = 2X \sim N(0, 4)$.
• Conclusion: The distribution of $X + Y$ has a discrete point mass at exactly 0 with a probability of 0.5, combined with a continuous normal curve for the remaining 0.5 probability. A true normal distribution cannot contain a discrete point mass, thus $X + Y$ is not normally distributed.